

William Tucker

Senior Software Engineer · AI Engineer · Kelowna, BC

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Professional Summary

Senior software engineer with **12+ years** designing and shipping backend systems, data pipelines, and full-stack applications, with current focus on AI-native delivery — direct **Anthropic Claude API** integrations, a custom **Oracle MCP server** authored for my team, and AI-augmented modernization shipping in days instead of months.

Strong in **C#/.NET, Oracle, REST APIs**, and **CI/CD**; experienced in legacy modernization (Classic ASP to .NET), ETL, and technical leadership. I use AI agents to accelerate delivery: the same modernization work that once took months now ships in days, with full test coverage and CI/CD. On my own time, I built dogmap.ca — a production PWA with Next.js, Supabase, PostGIS, OAuth, RLS, and JWT-protected API routes.

I care about infrastructure, security, and knowledge transfer. I document everything so teams own what gets built.

Core competencies: Backend & integrations · AI-accelerated development · Legacy modernization · CI/CD & DevOps · Security (SSO, OAuth, RLS, JWT) · Oracle & SQL · Cloud (Supabase, Vercel) · Technical leadership & documentation

Technical Skills

Category	Technologies
Languages	C#, Java, JavaScript, TypeScript, Python, SQL, PL/SQL, T-SQL, Groovy, VBScript, PowerShell, XML
Backend / APIs	.NET 8, ASP.NET Core, Blazor Server, Classic ASP, REST, SOAP, Dapper, Entity Framework, SignalR, Supabase

Data	Oracle, SQL Server, PostgreSQL, PostGIS (geospatial), stored procedures, schema design, ETL (Talend)
Frontend	Next.js, React, Blazor, TypeScript, TailwindCSS, Bootstrap 4/5, HTML/CSS, jQuery, MapLibre GL
AI & LLMs	Anthropic Claude API (direct SDK integration), GPT-5, Claude Code, custom MCP server authoring (TypeScript + <code>oracledb</code>), tool use / function calling, prompt engineering, AI-agent-driven development, context document design
Reporting	QuestPDF, Crystal Reports, PL/SQL, T-SQL
DevOps / Tools	Git (multi-repo, 30+ repos), GitLab CI/CD, Vercel, PowerShell, YAML, SSH/SFTP, multi-environment release (dev/test/prod)
Cloud	Vercel, Supabase (Auth, DB, Storage), cloud ERP integrations (REST/SOAP), env-based config
Testing	Unit testing, integration testing, code review
Security	SSO/ADFS, SAML, token validation, OAuth (Google, Apple), Row Level Security (RLS), JWT, CORS architecture, role-based access, rate limiting
Other	IIS, Windows Server, Grails, Agile/Scrum

Experience

AI Engineer & Independent Developer — William Tucker Solutions

2026 – Present · Kelowna, BC (Remote)

- **Built `williamtucker.ca` production AI chatbot** using the Anthropic Claude API (`@anthropic-ai/sdk` direct SDK integration) — custom system prompts, rate limiting, session management, Node.js/Express backend, deployed to Railway.
- **Built WTSAdmin + client portal** — Next.js 16 (App Router, Turbopack) + Supabase (PostgreSQL + Auth + Storage + Row-Level Security) + Stripe + Resend + React PDF; 9 admin pages, 5 client portal pages, magic-link auth, PDF invoicing, route protection via Next.js 16 `proxy.ts` . Sole author, 4 days end-to-end.

- **AI-augmented delivery practice** with Claude Code, Cursor, and GPT-5 — modernization work that previously took months now ships in days with CI/CD and automated tests; planning, debugging, and design decisions stay with me.
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Programmer/Analyst — Vancouver Island University

Jul 2016 – Present · Nanaimo, BC

- **Delivered a 3-project sprint in 12 days (2026-04-09 → 2026-04-20)** using parallel AI-agent sessions — greenfield .NET 8 Blazor SSO portal (50 commits; ~~standalone SAML login, impersonation with privilege check hardening, pen-test pass, CSP compliance~~), artifact-based SRS deploy pipeline (17 commits; Pester v3 test harness, pre-deploy snapshot retention, manual prod gate, Teams Adaptive Card notify), and a team-wide Obsidian documentation vault (~40 commits; Oracle SSO-chain schema reference, Classic ASP shared-includes reference, end-to-end SSO traces with mermaid diagrams). See [PROFILE.md](#) for full breakdown and independent git verification.
- **Led full modernization** of Facilities Information System from Classic ASP to .NET 8 Blazor Server — sole author, ~7 weeks (66 commits across 14 branches); preserved Oracle backend and cookie-based SSO; GitLab CI with test/build/deploy stages; architecture adopted as institutional conversion guide. Now in production.
- **Rebuilt room booking system** as a .NET 8 Blazor Server app — sole author, 5 days end-to-end (49 commits across 24 branches), replacing a legacy Oracle PL/SQL system (~21K LOC, 14K-line `findroom.sql`); mobile-responsive with real-time conflict detection. Currently in QA, technically ready for production; go-live pending organizational approval.
- **Built SSO landing page tag system** — designed Oracle schema (3 tables, sequences, triggers, cascading deletes), implemented 9 REST API endpoints in C#, and rewrote the entire landing page UI with modern patterns (slide-over panels, toast notifications, inline prompts replacing alert/confirm); includes JSON and Mermaid diagram export.
- **Authored a custom Oracle MCP server** in TypeScript (`@modelcontextprotocol/sdk` + `oracledb`) extending Claude Code with multi-database Oracle tooling — thick-mode LDAP TNS resolution, per-DB connection pools, read/write separation guardrails, row-limit enforcement;

wired into the team's shared `.mcp.json`. Daily driver against ODEV.
(github.com/billski/Claude-Oracle-MCP)

- **Replaced Crystal Reports with QuestPDF** in the Building Information System; eliminated legacy licensing dependency with a modern .NET-native reporting library.
- **Built CDW submission tool** — .NET web app with SignalR real-time progress, SSH/SFTP file transfer via SSH.NET, service account management, and GitLab CI pipeline with automated tests.
- **Established local development environment** for a team of six; transitioned from server-based development to local feature-branch workflow across 30+ git repositories; designed whitelist `.gitignore` strategy, multi-repo rollback procedures, and health-check-gated GitLab CI/CD pipelines across dev, test, and prod.
- **Authored modernization assessments** with cost-benefit analysis and phased conversion plans for legacy Classic ASP applications; produced institutional conversion guide used as a reference for future projects.
- **Implemented and documented** authentication and access control across web and .NET API services (SSO/ADFS, SAML, token validation, privilege checks); designed dynamic CORS architecture enabling multi-developer local development.
- **Designed team developer handbook** with stakeholder input; single source of truth for branching strategy, pipelines, and onboarding.
- **Developed integrations** from legacy systems to a modern cloud ERP using REST/SOAP in C# .NET and Entity Framework.
- **Delivered compliance work** — reviewed and updated SQL/XML for CRA T4A/T2202A electronic filing (2026 standards); resolved XML output and parsing issues for student and vendor extracts.
- **Participate in Agile/Scrum** ceremonies; contribute to code reviews and unit/integration testing.

Software Analyst — Thompson Rivers University

Jan 2013 – Jul 2016 · Kamloops, BC

- **Designed and built** the T.R.U. Student ID Card system in Java; used by thousands of students, integrated with BC Transit buses and City of Kamloops.

- **Created** Employee Survey system on Groovy Grails stack with Oracle; comprehensive reports and ETL jobs across Finance, HR, Payroll, and Student departments.
 - **Delivered** Groovy Grails presentation at BCNET Conference 2015; trained staff on institutional reporting, code review, and SQL tuning.
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Institutional Data Analyst — Thompson Rivers University

Sep 2012 – Mar 2013 · Kamloops, BC

- Reporting on Oracle data using SQL and PL/SQL.
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Institutional Report Coordinator (Co-op) — Thompson Rivers University

May 2012 – Dec 2012 · Kamloops, BC

- SQL, PL/SQL, and PHP development for institutional reporting.
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Education

Degree	Institution · Years
Computer Science Diploma	Thompson Rivers University · 2010 – 2013
	<i>Computer Systems: Operations & Management, Computer Science</i>
Journeyman Marine Technician	British Columbia Institute of Technology · 2003 – 2007

Certifications

Certification	Issuer	Date
Java and Hibernate	TestDome	Aug 2021
SQL (Intermediate)	HackerRank	Aug 2021

Key Projects

Project	Outcome
Facilities Information System modernization	Classic ASP → .NET 8 Blazor Server; same Oracle schema and SSO; sole-author ~7 weeks (66 commits, 14 branches); in production; reference implementation for institutional conversion guide.
Room booking system	Legacy Oracle PL/SQL room booking system (~21K LOC, 14K-line <code>findroom.sql</code>) → .NET 8 Blazor Server; sole author, 5 days end-to-end; mobile-responsive, real-time conflict detection; in QA, blocked on org approval.
SSO landing page + tag system	Rewrote SSO landing page UI; designed and built database-backed tag/layer system (Oracle schema, 9 REST endpoints, 12 SQL migrations, JSON/Mermaid export); modern UI with slide-over panels and toast notifications.
Crystal Reports → QuestPDF	Replaced legacy licensed reporting in BIS with .NET-native QuestPDF; eliminated licensing dependency.
Local dev environment & CI/CD	Transitioned 30+ repos from server-based to local feature-branch development; GitLab pipelines with health-check gates, rollback automation, and multi-environment release.
Developer handbook	Designed with team input; central resource for workflow, branching, pipelines, and onboarding.
dogmap.ca (side project)	PWA for BC dog owners. Next.js, TypeScript, Supabase (PostgreSQL, PostGIS, Auth, Storage), Vercel. RLS on all tables, OAuth (Google, Apple), JWT in API routes, role-based access, admin dashboard, moderation queue. Live at dogmap.ca .
williamtucker.ca	Production website with live AI chatbot built directly on the Anthropic Claude API (<code>@anthropic-ai/sdk</code>) — custom system prompts, rate limiting, session management, Node.js/Express backend; Tailwind CSS, deployed to Railway.
WTSAdmin + client portal	Full-stack SaaS shipped in 4 days, sole author. Next.js 16 (App Router, Turbopack) + Supabase (PostgreSQL + Auth + Storage + RLS) + Stripe + Resend + React PDF. 9 admin pages, 5 client portal pages, magic-link auth, PDF invoicing, route protection via Next.js 16 <code>proxy.ts</code> .

Oracle MCP server	Custom MCP server in TypeScript (<code>@modelcontextprotocol/sdk</code> + <code>oracledb</code>) giving Claude Code first-class Oracle tools across multiple databases — thick-mode LDAP, multi-DB connection pools, read/write separation, row-limit guardrails. Authored in a day, daily driver against ODEV. github.com/billski/Claude-Oracle-MCP
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